METRIC

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SUPERSEDING

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PERFORMANCE SPECIFICATION

TRANSIENT VOLTAGE SURGE SUPPRESSOR (TVSS)

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE.

1.1 Scope. This performance specification covers the requirements for a transient voltage surge suppressor (TVSS) device used to provide protection from electrical system transients for electrical equipment installed aboard U.S. Navy vessels and land-based Navy electrical distribution systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/26307	-	Gray, Semigloss

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-E-917	-	Electric Power Equipment Basic Requirements
MIL-DTL-15090	-	Enamel, Equipment, Light Gray, (Navy Formula No. 111)
MIL-DTL-24643		Cables and Cords, Electric, Low Smoke, for Shipboard Use, General Specification for
MIL-DTL-24643/17		Cable, Electrical, 1000 Volts, Type LSFSGU

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1310	-	Standard Practice for Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety
MIL-STD-1399-300	-	Interface Standard for Shipboard Systems, Section 300, Electric Power, Alternating Current (Metric)
DOD-STD-1399-301	-	Interface Standard for Shipboard Systems, Section 301, Ship Motion and Attitude (Metric)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM F1507	-	Standard Specification for Surge Suppressors for Shipboard Use

(Copies of this document are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 45	-	IEEE Recommended Practice for Electric Installations on Shipboard (DoD adopted)
IEEE C62.41.1	-	IEEE Guide on the Surge Environment in Low-Voltage (1000 V and less) AC Power Circuits
IEEE C62.41.2	-	IEEE Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and less) AC Power Circuits
IEEE C62.45	-	IEEE Recommended Practice on Surge Testing for Equipment Connected to Low-Voltage (1000V and Less) AC Power Circuits

(Copies of these documents are available from the Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331 or online at www.ieee.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA 250	-	Enclosures for Electrical Equipment (1000 Volts Maximum) (DoD adopted)
NEMA LS-1	-	Low Voltage Surge Protective Devices

(Copies of these documents are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209 or online at www.nema.org.)

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UNDERWRITERS LABORATORIES, INC. (UL)

UL 94	-	Tests for Flammability of Plastic Materials for Parts in Devices and Appliances (DoD adopted)
UL 1449	-	Transient Voltage Surge Suppressors (DoD adopted)

(Copies of these documents are available from COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515 or online at www.ul.com.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a TVSS device shall be subjected to first article inspection in accordance with 4.2.

3.2 Description. A TVSS device described in this specification is an electronic device for low voltage (<600 Vrms) systems that is added in parallel to an electrical circuit to provide protection to the load from transient voltages in the electrical distribution system. This is accomplished by attenuating or reducing, to an acceptable level, the magnitude of electrical distribution system disturbances such as transient voltage levels and noise that may be seen by the load. The TVSS device shall be placed at interfaces between shipboard electrical power distribution systems and individually connected user equipment and distribution panels. The TVSS device shall be suitable for use on existing shipboard and land-based Navy electrical distribution system.

3.3 Electrical characteristics. The TVSS device shall operate within the Naval electrical distribution system parameters specified by MIL-STD-1399-300. The TVSS device shall perform in the category B surge location described in IEEE C62.41.1, IEEE C62.41.2, and IEEE C62.45.

3.3.1 Circuit design. A TVSS device may consist of, but not be limited to, components such as insulated gate bipolar transistors (IGBTs), metal oxide varistors (MOVs), silicon avalanche diodes (SADs), gas discharge tubes (GDTs), and passive filter components or any combination thereof as a hybrid. The TVSS device shall be one port, Class B, Type I in accordance with ASTM F1507. Lead lengths of TVSS devices shall be as short as possible in order to ensure optimum operation (see 3.3.9). The design and construction shall be in accordance with MIL-E-917.

3.3.2 Device circuit configuration. The TVSS device shall be available in either single-phase or three-phase delta or wye configuration as specified (see 6.2).

3.3.2.1 Single-phase. The single-phase configuration shall be constructed for use with two hot wires and an equipment ground wire.

3.3.2.2 Three-phase. The three-phase configuration shall be delta or wye.

3.3.2.2.1 Delta. The delta configuration shall be constructed with three-phase wires and an equipment (chassis) ground wire.

3.3.2.2.2 Wye. The wye configuration shall be constructed with three-phase wires, a neutral wire, and an equipment (chassis) ground wire.

3.3.3 Maximum continuous operating voltage (MCOV). The MCOV for the TVSS device shall be as specified in Table I for rigorous Navy use using ASTM F1507 and NEMA LS-1 as guides for line-to-line (L-L) and line-to-ground (L-G). Verification that MCOV ratings are met shall be provided as specified (see 6.2).

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TABLE I. MCOV ratings.

System voltage	Single-phase device		3-phase delta device		3-phase wye device		
L-L	L-L	L-G	L-L	L-G	L-L	L-N	L-G
115 Vrms	≥150 Vrms	≥150 Vrms	≥150 Vrms	≥150 Vrms	≥150 Vrms	≥150 Vrms	≥150 Vrms
200 Vrms	≥260 Vrms	≥260 Vrms	≥260 Vrms	≥260 Vrms	≥260 Vrms	≥260 Vrms	≥260 Vrms
440 Vrms	≥550 Vrms	≥550 Vrms	≥550 Vrms	≥550 Vrms	≥550 Vrms	≥550 Vrms	≥550 Vrms

3.3.4 Minimum peak current per mode. The minimum peak current per mode of the TVSS device shall be 40,000 A or greater with a 8 x 20 microsecond waveshape. The protected modes shall consist of all line-to-line, line-to-ground, line-to-neutral, and neutral-to-ground configurations. The phases are identified as A, B, and C. For a delta configuration, the modes are defined as AB, AG, BC, BG, CA, and CG. For a wye configuration, the modes are defined as AB, AG, AN, BC, BG, BN, CA, CG, CN, and NG. Verification that the minimum peak current per mode of the TVSS device is met shall be provided as specified (see 6.2).

3.3.5 Suppressed voltage ratings (SVRs). The maximum SVRs shall be as shown in Table II from L-L, L-N, and L-G at the system voltage for delta and wye systems. Testing shall be performed as in 4.5.12 with results consistent with UL 1449 SVRs per UL 1449 product test report. When specified (see 6.2), a copy of the UL 1449 test report shall be provided.

TABLE II. Maximum allowable UL 1449 suppressed voltage ratings.

System voltage	Single-phase device		3-phase delta device		3-phase wye device		
L-L	L-L	L-G	L-L	L-G	L-L	L-N	L-G
115 Vrms	500 V	500 V	500 V	500 V	500 V	500 V	500 V
200 Vrms	900 V	900 V	900 V	900 V	900 V	900 V	900 V
440 Vrms	1500 V	1500 V	1500 V	1500 V	1500 V	1500 V	1500 V

3.3.6 Maximum let-through voltage ratings. The maximum let-through voltage ratings shall be as shown in Table III from line-to-line (L-L), line-to-neutral (L-N), and line-to-ground (L-G) at the system voltage for delta and wye systems. Testing shall be performed as in 4.5.5 with results consistent with Table III.

TABLE III. Maximum allowable let-through voltages.

System voltage	Single-phase device		3-phase delta device		3-phase wye device		
L-L	L-L	L-G	L-L	L-G	L-L	L-N	L-G
115 Vrms	675 V	675 V	675 V	675 V	675 V	675 V	675 V
200 Vrms	900 V	900 V	900 V	900 V	900 V	900 V	900 V
440 Vrms	1750 V	1750 V	1750 V	1750 V	1750 V	1750 V	1750 V

3.3.7 Input power frequency. The TVSS device shall be fully operational at the system frequencies of 60 and 400 Hz. This shall be verified by the requirements of 4.5.4.

3.3.8 Protected modes. Modes as defined in 3.3.4 shall be protected using directly connected suppression circuit elements. For a wye configured TVSS device, the elements shall be connected across L-L, L-N, L-G, and N-G modes. For a delta-configured TVSS device, the elements shall be connected across L-L and L-G modes.

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3.3.9 Circuit protection. Each line connection from the distribution system phase conductor to the TVSS device shall be fused and properly coordinated to disconnect the TVSS device from the distribution system in the event of a TVSS failure. Individual fuses, such as VSP-type surge suppressor protection fuses, one per phase, internal to the TVSS enclosure, shall be sized to protect the feeder cable. Appropriate feeder cable such LSFSGU-9, specified in MIL-DTL-24643 and MIL-DTL-24643/17, not to exceed 18 inches in length, shall be used. The cable can then be sized as short as is practical for the installation.

3.3.10 Status indication. The TVSS device shall provide a visual indication of fuse status per phase. When power is applied and the fuse is closed, the indication shall be lit; if the fuse opens, the indication shall extinguish.

3.3.11 Grounding. The TVSS device enclosure shall provide a connection point to facilitate equipment grounding in accordance with MIL-STD-1310.

3.3.12 UL 1449 certification. The TVSS device shall be UL 1449 listed as having undergone UL 1449 performance tests (including SVR testing) for UL 1449 certification. Verification of this requirement shall be provided as specified (see 6.2).

3.3.13 Life-cycle. The TVSS device shall be able to withstand 2500 each L-L and 2500 each L-G 1.2 x 50 ms waveshape at 1 kV magnitude for 115 Vrms devices and 2.5 kV for 440 Vrms devices per MIL-STD-1399-300 at a frequency of 2 impulses per minute. The TVSS device shall withstand such testing with zero failures and less than 10 percent degradation of let-through voltage parameter.

3.4 Mechanical characteristics.

3.4.1 Enclosure requirements. The TVSS device components shall be mounted in a NEMA 4-hinged enclosure with a seamless weld in accordance with NEMA 250, unless otherwise specified. The mounting shall meet the requirements of IEEE 45 The overall size of the enclosure shall be minimized to meet application requirements. When specified (see 6.2), non-magnetic enclosures are required for minesweeper ships.

3.4.2 Component mounting. No components other than indicator lights shall be mounted external to the enclosure.

3.4.3 Dimensions and weight. The maximum dimensions for the enclosure shall be 30 cm (12 in) in height by 30 cm (12 in) in width by 30 cm (12 in) in depth, inclusive of all mounting hardware. The TVSS device shall weigh less than 18.144 kg (40 lbs).

3.4.4 Painting. The TVSS device enclosure shall be painted using Navy formula no. 111 in accordance with MIL-DTL-15090. The color shall be no. 26307 in accordance with FED-STD-595.

3.4.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.6 Identification plates. The TVSS device shall be identified with a label plate that shall include the following information:

a. Manufacturer’s name, address, and phone number
b. Model and serial number
c. Nominal operating voltage and frequency
d. Maximum continuous operating voltage
e. Maximum surge current in kA for an 8/20 microsecond waveshape
f. Intended power system configuration
g. Underwriters Laboratories listing data
h. Installation instructions
i. Warranty information

It is understood that more than one label may be utilized to provide all information specified above.

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3.4.7 Encapsulant. Any epoxy or encapsulant shall meet the requirements of UL 94 for reduced flammability. Additionally, the Hazardous Materials Indication System (HMIS) value of the cured encapsulant shall be zero.

3.5 Environmental conditions.

3.5.1 Location. The TVSS device shall be installed in a sheltered location aboard the ship; as such, the device will be protected from the external environment. The sheltered environment may be either controlled or uncontrolled.

3.5.2 Ambient temperature. The TVSS device shall withstand storage temperatures of -25 to 70 °C (-13 to 158 °F) and ambient operating temperatures of -25 to 70 °C (-13 to 158 °F) without any derating of the device.

3.5.3 Relative humidity. The TVSS device shall operate in a relative humidity of 0 to 100 percent (non-condensing).

3.5.4 Corrosion. The TVSS device shall be corrosion-resistant in the marine environment in accordance with NEMA 250.

3.5.5 Inclination. The TVSS device shall operate at any inclination.

3.5.6 Ship motion and attitude. The TVSS device shall be fully operational for the ship motion and attitude conditions set forth in DOD-STD-1399-301.

3.5.7 Shock. The TVSS device shall be classified as Grade A, Class I, Type A in accordance with MIL-S-901. It shall withstand the shock loads as specified in MIL-S-901. Resilient shock mounts may be used. The TVSS device shall fail the shock testing if any of the following occur:

a. A portion of the device or enclosure breaks or causes a hazard to personnel.

b. Wire connections, terminal strips, and other device components crack or become dislodged.

c. Any part breaks or cracks that affects device function.

d. There is appreciable distortion of any parts, including enclosure and framework.

e. The device fails to pass visual examination.

f. The device fails to pass insulation resistance tests.

3.5.8 Vibration. The TVSS device shall be in accordance with the Type I vibration requirements of MIL-STD-167-1. It shall withstand the vibration testing as specified in MIL-STD-167-1. Resilient vibration mounts may be used. The TVSS device shall fail the vibration testing if any of the following occur:

a. A portion of the device or enclosure breaks or causes a hazard to personnel.

b. Wire connections, terminal strips, and other device components crack or become dislodged.

c. Any part breaks or cracks that affects device function.

d. There is appreciable distortion of any parts, including enclosure and framework.

e. The device fails to pass visual examination.

f. The device fails to pass insulation resistance tests.

3.6 Operation.

3.6.1 Duty cycle. The duty cycle of the TVSS device shall be continuous operation.

3.7 Maintenance. The TVSS device shall require no preventive maintenance.

3.8 Performance characteristics. The TVSS device shall conform to the requirements of ASTM F1507, and shall be listed by Underwriters Laboratories, Inc. under UL 1449.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article inspection shall be performed on one of each model/configuration TVSS device when a first article sample is required (see 3.1). The inspection shall include the examination specified in 4.4 and all tests specified in Table IV. See 4.2.1 for sequence of testing.

4.2.1 Sequence of testing. Testing shall be performed on the UL listed TVSS device in the following sequence:

a. Operation test (see 4.5.1).

b. Assembly test (see 4.5.2).

c. Workmanship test (see 4.5.3).

d. 400-Hz operational (MCOV) test (see 4.5.4).

e. Grounding continuity test as performed in UL 1449 (see 4.5.12).

f. Vibration testing (see 4.5.7).

g. Grounding continuity test as performed in UL 1449 (see 4.5.12).

h. Shock testing (see 4.5.8).

i. Grounding continuity test as performed in UL 1449 (see 4.5.12).

j. Temperature testing (see 4.5.9).

k. Grounding continuity test as performed in UL 1449 (see 4.5.12).

l. Let-through voltage testing (see 4.5.5).

m. Life cycle testing (see 4.5.6).

n. Let-through voltage testing (see 4.5.5) to verify not more than a 10 percent difference from the previous let-through voltage testing.

o. Minimum peak current testing (see 4.5.11).

p. Grounding continuity test as performed in UL 1449 (see 4.5.12).

4.3 Conformance inspection. The conformance inspection shall consist of the examination specified in 4.4 and the tests specified in Table IV.

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TABLE IV. Tests.

Test	Requirements	First article test method	Conformance test method
Operation	3.3.10	4.5.1	4.5.1
Assembly	3.3.2, 3.3.10	4.5.2	4.5.2
Workmanship	3.3.2, 3.3.10	4.5.3	4.5.3
400-Hz operational test	3.3.7	4.5.4	N/A
Let-through voltage testing	3.3.6	4.5.5	N/A
Life cycle tests	3.3.13	4.5.6	N/A
Vibration tests	3.5.8	4.5.7	N/A
Shock tests	3.5.7	4.5.8	N/A
Temperature tests	3.5.2	4.5.9	N/A
Maximum continuous operating voltage (MCOV)	3.3.3	4.5.10	N/A
Minimum peak current per mode	3.3.4	4.5.11	N/A
UL 1449 certification	3.3.12	4.5.12	N/A

4.4 Examination. Each TVSS device shall be visually examined for compliance with the requirements of 3.4. At the contractor's option, automated optical and electronic sensors may perform portions of the visual examination.

4.4.1 Dimensions. External dimensions shall be measured for conformance with the requirements of 3.4.3 and the size of bolt holes verified for mounting the units. The dimensions shall be measured and compared to the dimensions specified in this specification.

4.5 Methods of inspection.

4.5.1 Operation. Every production TVSS device shall be powered at the factory to demonstrate that installed indicator lights and fusing are operational. Characteristics of the indicator lights are described in 3.3.10.

4.5.2 Assembly. The TVSS device shall be visually inspected to ensure that all specified components are present. Where automated processes are employed for the assembly of TVSS devices, electro-optical devices may be used to perform the inspection.

4.5.3 Workmanship. Components shall be visually inspected for defects.

4.5.4 400-Hz (MCOV) operational test. The TVSS device shall be placed in a temperature-controlled chamber with an ambient temperature of 70 °C ±5 °C and the rated maximum continuous operating voltage at 400 Hz shall be applied for a period of 100 hours. The suppressor leakage currents shall be measured at the beginning of the test after the TVSS device temperature has stabilized and again after 100 hours. The leakage currents at the conclusion of the test shall be less than 30 mA and shall be less than 110 percent of the initial leakage current for the TVSS device.

4.5.5 Let-through voltage testing. The maximum let-through voltage shall be as shown in Table III for line-to-line (L-L) and line-to-ground (L-G), with a 6-inch lead length when subjected to a 6 kV/3 kA Location Category B impulse waveform as defined in IEEE C62.41.2. The oscilloscope used shall have a minimum bandwidth of 100 MHz and a minimum input impedance of 1 Mohm, capable of displaying the entire response of the TVSS device. Impulses shall be applied/measured at positive polarity at 90° voltage phase angle. Let-through voltages shall be measured from the positive peak of the sine wave to the positive peak of the impulse.

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4.5.6 Life cycle tests. The number of applied voltage impulses to the TVSS device shall be 2500 each L-L and 2500 each L-G; the impulses shall have a 1.2 x 50 microsecond waveshape and 1 kV magnitude for 115 Vrms devices and 2.5 kV for 440 Vrms devices per MIL-STD-1399-300, applied at 90 electrical degrees positive polarity of the steady state AC waveform. Impulses shall be applied on the order of 2 per minute. The TVSS device shall withstand life-cycle testing with zero failures and less than 10 percent degradation of let-through voltage parameter. Should any failures occur, corrective action shall be taken and the life cycle test restarted from the beginning. The TVSS device must endure the application of at least 5000 impulses (2500 line-to-line and 2500 line-to-ground) without failure.

4.5.7 Vibration tests. For conformance with the requirements of 3.5.8, a vibration response investigation shall be performed. The TVSS device shall be subjected to endurance testing in accordance with MIL-STD-167-1.

4.5.8 Shock tests. Conformance with the requirements of 3.5.7 shall be verified by testing in accordance with MIL-S-901.

4.5.9 Temperature tests. Temperature tests for conformance with the requirements of 3.5.2 shall be in accordance with MIL-STD-810, methods 501 and 502.

4.5.10 MCOV test. This test is satisfied by successfully passing testing performed in accordance with 4.5.4.

4.5.11 Minimum peak current per mode test. There is no AC sinusoidal power applied during this test process. All modes of the TVSS device shall undergo testing individually. All test waveforms are defined in IEEE C62.41.2. A particular mode shall be subjected to three 6 kV/3 kA (1.2/50 µs, 8/20 µs - combination waveform) pre-strikes to document baseline let-through voltages. The same mode then shall be subjected to a single 40,000 A 8/20 µs current waveform. Provisions shall be made to allow replacement of fusing that operates during the 80,000 A waveform. (This excludes replacing fusing during the pre- and post-strike tests at 6kV/3kA.) Finally, the same mode shall be subjected to another three 6 kV/3 kA post-strikes. Following the application of each test surge, the TVSS device shall be examined to determine if it suffered any damage. The surge current and let-through voltage plots shall be recorded.

4.5.12 UL 1449 certification. Certification shall be provided as specified (see 6.2) that the TVSS device is UL 1449 listed as having undergone UL 1449 performance tests (including SVR testing) for UL 1449 certification at 60 Hz (see 3.3.12).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The TVSS device specified herein is intended for the protection of sensitive shipboard electrical and electronic equipment. It is not intended to protect purely resistive loads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Specific issue of documents referenced (see 2.2).

c. When first article is required (see 3.1).

d. Circuit configuration requirements (see 3.3.2).

e. Verification that MCOV ratings are met (see 3.3.3).

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f. Verification that the minimum peak current per mode of the TVSS device is met (see 3.3.4).

g. When the UL 1449 test report is required (see 3.3.5).

h. Verification that the TVSS device is UL 1449 listed as having undergone UL 1449 performance tests (including SVR testing) for UL 1449 certification (see 3.3.12).

i. When non-magnetic enclosures are required for minesweeper ships (see 3.4.1).

j. Packaging requirements (see 5.1).

k. Shelf-life requirements (see 6.3).

6.3 Shelf-life. This specification covers items where shelf-life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DOD 4140.27-M, Shelf-life management Manual, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/.

6.4 Subject term (key word) listing.

Circuit protection

Electrical equipment

Electrical protection

Electronics

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - MI
Navy - SH
Air Force - 11

Preparing Activity:
Navy - SH
(Project 5920-0855-000)

Review Activities:
Army - CR
Navy - AS, CG
Air Force - 03, 19, 99
DLA - CC

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